

Fachhochschule Vorarlberg GmbH		Author: Horatiu O. Pilsan	
EIT & MEC - Digi SGL 1. Topic: Switching Circuits	Rev. 1.1	15.04.2026	Page 1 of 1

1. Objective:

Understand the mode of operation of switching circuits and get to know the different types.

2. Questions:

1. How does an astable multivibrator built with transistors work?
The same question for a monostable multivibrator
2. Understand the mode of operation of an astable and a monostable multivibrator built using the NE555.
3. Understand the mode of operation of a Pierce quartz oscillator.
Which are the pros and cons of quartz crystals compared to R-C (or L-C) circuits?

4. Tasks:

1. Sum up the most important pieces of information on the topic on one flip chart without adding too many details! Best suited is a visual manner.
2. Formulate questions on the topic, which you were not able to answer, neither alone nor in the team meeting!

5. Dates:

See course schedule in the information file for the course (_EIT-Digi_Info.pdf).

6. Literature:

- [1] Tietze, Ulrich; Schenk, Christoph (1999): Halbleiter-Schaltungstechnik, 11. Ausgabe, Berlin: Springer, (Kap. 6.2, p. 598 – 620)
- [2] Data sheet of the NE555 in ILIAS (folder „Lab Datasheet“), p. 9 – 11.
- [3] Reinhold, Wolfgang (2010): Elektronische Schaltungstechnik, München: Hanser (p. 295 – 297)
- [4] Oehme, Friedrich; Huemer, Mario; Pfaff Markus (2011): Elektronik und Schaltungstechnik. Ein verständlicher Einstieg, München: Hanser (p. 263 – 269).